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Research Profile

Graduate researcher, educator, and NSF Robert Noyce Scholar working at the intersection of artificial intelligence, data science, inclusive computer-science education, and reproducible experimentation. Current work includes fairness-aware sequential decision systems, quantum-network routing, healthcare analytics, and classroom uses of AI that preserve human judgment. More than ten years of engineering and data-systems experience support the translation of research ideas into reliable tools, learning activities, and documented evaluation workflows.

Research Interests

AI for education; learning-science-informed intelligent systems; fairness-aware machine learning; human-centered evaluation; neurodiversity and accessible computing education; reproducible benchmarking; sequential decision-making; quantum-classical hybrid methods.

Education

Rochester Institute of Technology Expected Dec 2026
Master of Science, Data Science — GPA 3.9 Rochester, NY

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Foundations of Data Science (DSCI-633), Software Engineering for Data Science (DSCI-644), Applied Data Science I (DSCI-601), Bioinformatics Algorithms (BIOL-630), Data-Driven Knowledge Discovery (ISTE-780).

University of Rochester, Warner School of Education Expected Dec 2026
Master of Science, Teaching Computer Science K-12 — GPA 4.0 Rochester, NY

- NSF Robert Noyce Teacher Scholarship Program (full tuition waiver).
- Advanced Certificates in Teaching Students with Disabilities, All Grades, and Digitally-Rich Teaching in K-12 Schools (in progress).

Rochester Institute of Technology 2015
Master of Science, Computer Science — GPA 3.2 Rochester, NY

- Concentrations: Computer Graphics, Artificial Intelligence, Big Data Analytics.
- Coursework in compiler design, algorithm analysis, and distributed systems.

Farmingdale State College 2012
Bachelor of Science, Computer Engineering Technology — GPA 3.3 Farmingdale, NY

- Dual degree: Electrical Engineering & Computer Engineering Technology.
- Honors: CSTEP Award & Merit, Dual BS Scholarship.

Research and Professional Experience

Rochester Institute of Technology Fall 2025 – Present
Graduate Assistant, Quantum and AI Research Rochester, NY

- Build a Python multi-testbed framework for adaptive quantum routing and qubit allocation using multi-armed, contextual, adversarial, and neural bandit methods.
- Design reusable feedback, validation, logging, dataset, and experiment-tracking workflows across stochastic and adversarial environments.
- Evaluate utility, latency, reliability, robustness, and group-level outcomes rather than relying on one aggregate score.

University of Rochester / Pine Brook Elementary School Feb 2026 – May 2026
Computer Science Student Teacher Rochester, NY

- Co-designed and delivered K-5 computer science instruction using computational thinking, inclusive pedagogy, Universal Design for Learning, and multiple participation formats.
- Produced lesson artifacts and instructional analyses connecting AI literacy, accessible education, and evidence-based computer-science pedagogy.

Rochester Institute of Technology

Equitable Bioinformatics Research — ISTE-780

2024 – 2025

Rochester, NY

- Applied systematic fairness analysis to RNA secondary structure prediction pipelines; demonstrated statistical significance ($p < 0.01$) in bias detection across algorithmic variants using SHAP and disparity-impact analysis.
- Developed reproducible evaluation codebase (900+ lines) with automated fairness-metric reporting and statistical testing.

Rochester Institute of Technology

Computational Biology Research (BIO614 and BIO550)

2025 – 2026

Rochester, NY

- Led formal computational analysis in BIO614: advanced RNA secondary structure prediction with thermodynamic MFE enhancements; produced *BIO614-FinalProjectProposal* (Overleaf manuscript).
- Conducted reproducible bulk RNA-seq differential-expression analysis (BIO550): DESeq2 pipeline with QC validation and biological interpretation of nociceptor gene expression.

VEDADATA Inc.

Data Solutions Engineer

Sep 2022 – Aug 2024

New York, NY

- Designed scalable Python workflows for ingestion, validation, and statistical diagnostics, emphasizing data reliability and structured quality checks.
- Strengthened production reliability through anomaly detection, schema validation, and structured logging.

VIOME Inc.

AI Data Solutions Engineer

Jan 2021 – Sep 2022

New York, NY

- Developed healthcare machine-learning workflows covering preprocessing, feature engineering, supervised experimentation, evaluation, and deployment-oriented data preparation in AWS-based environments.

Selected Technical Writing

- *Big Data Medical Diagnosis: A Multi-Method Approach* (RIT M.S. Computer Science graduate research, 2015) — ensemble and deep learning methods for clinical classification on high-dimensional biomedical data.
- *Predicting Hospital Readmission Rates: A Multi-Method ML Analysis* (DSCI-633 Project Report, 2025) — ensemble and regularized regression methods for clinical risk stratification under distribution shift.
- *BIO614-FinalProjectProposal* (Overleaf manuscript, 2026) — RNA secondary structure prediction with thermodynamic deep learning enhancements, reproducible benchmarking, and biological interpretation.
- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report, 2025) — fairness-aware ML for genomic diagnostic algorithms with SHAP-based bias detection and statistical significance testing.
- *Differential Gene Expression in Murine DRG Neurons Following Sciatic Nerve Injury: An NGS Reanalysis* (BIO550 Computational Biology Project, 2026) — reproducible bulk RNA-seq pipeline using DESeq2, QC validation, and pathway-level biological interpretation of nociceptor gene expression.
- *Fairness-Aware Bandits for Clinical Decision Systems* (DSCI-601 Project Proposal, 2025) — bandit learning with equity constraints applied to high-stakes clinical diagnostic decision-making, addressing demographic disparity in sequential clinical recommendation workflows.
- *Quantum Network Path Optimization and Fault-Tolerant Routing* (research manuscript, 2025–2026) — quantum-classical hybrid ML for fault-tolerant routing under noise constraints and decoherence.

Awards & Recognition

- NSF Robert Noyce Teacher Scholarship (2024–2026) — full tuition waiver for M.S. in Teaching Computer Science K–12.
- MS International Scholarship (2012–2015) — MESCYT & RIT, awarded for academic excellence.
- Honorable Mention Technology Award (2012) — NYS STEP/CSTEP, 2nd place capstone project.
- IEEE Alumni Member (2009–Present) — Member No. 90613000.

Technical Competencies

Programming: Python, R, Java, C++, C#, MATLAB, SQL, JavaScript, Bash

Research Systems: Python multi-testbed evaluation, contextual and adversarial bandits, validation, logging, dataset

and experiment tracking

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, fairness-aware ML, contextual bandit workflows

Data & Analysis: pandas, NumPy, SciPy, SHAP analysis, statistical testing, reproducible benchmarking

Infrastructure: AWS, GCP, Docker, Git/GitHub, Linux, LaTeX

Selected Fit for AI for Education Research

- Combines current AI and sequential-decision research with formal K–12 computer-science teacher preparation and direct elementary, middle-school, and environmental-science teaching experience.
- Designs accessible AI-supported learning activities using backward design, Universal Design for Learning, multimodal participation, and explicit routines for questioning and revising model outputs.
- Builds reproducible Python-based experimental and evaluation workflows across education, healthcare, bioinformatics, and quantum-network domains.
- Brings a disability-informed and cross-cultural perspective to research on how intelligent systems can support learning without replacing educators, peer exchange, or student agency.